Section B

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
8.	*(a)*			Fe_2O_3 + 3CO $\rightarrow$ 2Fe + $3CO_2$ (1) Any of the following for (1) • During this process the iron(III) ions (in Fe_2O_3) gain electrons (to produce iron); reduction is a process of electron gain • The oxidation number of iron is reduced from +3 to 0; a reduction in (positive) oxidation number is reduction • Carbon monoxide loses electrons; oxidation is a process of electron loss • The oxidation number of carbon is increased from +2 to +4; an increase in (positive) oxidation number is oxidation • Fe_2O_3 loses oxygen and CO gains oxygen	2			2		
	(b)			350 tonnes of which 0.02 % is sulfur $\therefore$ Mass of sulfur $= \dfrac{350 \times 0.02}{100} =$ 0.07 tonnes (1) Mg + S $\rightarrow$ MgS 0.07 tonnes sulfur needs $\dfrac{24.3 \times 0.07}{32.1} =$ 0.0530 tonnes = 53.0 (kg) (1) ecf possible	2			2	1	
	(c)			Cubic structure shows alternating different ions (1) Ions labelled as Mg^{2+} and S^{2-} (1)		2		2		